



Operations Research

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Solution to the Training Exercises, Sheet 4

Training Exercises

Exercise T4.1. *Duality & Complementary Slackness*

Problem

- a) Derive the dual LP from the following LP.

$$\begin{aligned} (P) \quad & \underset{\text{s.t.}}{\text{Min}} && 2x_1 + 2x_2 + 2x_3 - x_4 \\ & && -x_1 + x_2 + 2x_3 - x_4 \geq 15 \\ & && x_1 - x_2 + 4x_4 = 5 \\ & && 2x_2 - 2x_3 + x_4 \leq -15 \\ & && x_1, x_2, x_3 \geq 0 \\ & && x_4 \leq 0 \end{aligned}$$

- b) Let $x^T = (5, 0, 10, 0)$ be a primal point. Use the complementary slackness conditions to check whether x is an optimal solution for (P) .

Solution

- a)

$$\begin{aligned} (D) \quad & \underset{\text{s.t.}}{\text{Max}} && 15y_1 + 5y_2 - 15y_3 \\ & && -y_1 + y_2 \leq 2 \\ & && y_1 - y_2 + 2y_3 \leq 2 \\ & && 2y_1 - 2y_3 \leq 2 \\ & && -y_1 + 4y_2 + y_3 \geq -1 \\ & && y_1 \geq 0 \\ & && y_3 \leq 0 \end{aligned}$$

- b) The complementary slackness conditions are given by $(Ax - b)^T y = 0$ (primal) and $(A^T y - c)^T x = 0$ (dual), i.e.,

$$\begin{array}{ll}
(-x_1 + x_2 + 2x_3 - x_4 - 15)y_1 = 0 & (-y_1 + y_2 - 2)x_1 = 0 \\
(x_1 - x_2 + 4x_4 - 5)y_2 = 0 & (y_1 - y_2 + 2y_3 - 2)x_2 = 0 \\
(2x_2 - 2x_3 + x_4 + 15)y_3 = 0 & (2y_1 - 2y_3 - 2)x_3 = 0 \\
& (-y_1 + 4y_2 + y_3 + 1)x_4 = 0
\end{array}$$

First, we check if x is feasible by plugging it into (P) . We get $[15 \geq 15], [5 = 5], [-20 \leq -15], [10 \geq 5 \geq 0]$. Since $5, 0, 10 \geq 0$ and $0 \leq 0$, x satisfies all constraints and is therefore a feasible point.

Since the third primal constraint has slack, we have $y_3 = 0$. Furthermore, since $x_1, x_3 \neq 0$ we have $(-y_1 + y_2 - 2) = (2y_1 - 2y_3 - 2) = 0$. Thus, we get $y_1 = 1$ and $y_2 = 3$.

Finally, we check the feasibility of the derived point $y^T = (1, 3, 0)$. We get $[2 \leq 2], [-2 \leq 2], [2 \leq 2], [11 \geq -1]$ and $1 \geq 0, 0 \leq 0$. Thus y is feasible and by the complementary slackness theorem x has to be an optimal solution of (P) .

Exercise T4.2. Duality

Problem

Consider the following linear programs, with $A \in \mathbb{R}^{n \times n}$ and $b, c \in \mathbb{R}^n$.

$$\begin{array}{ll}
(P_1) \max_{\text{s.t.}} & c^T x \\
& Ax = b \\
& x \in \mathbb{R}_{\geq 0}^n \\
(P_3) \min_{\text{s.t.}} & b^T x \\
& A^T x \geq c \\
& x \in \mathbb{R}^n
\end{array}
\quad
\begin{array}{ll}
(P_2) \min_{\text{s.t.}} & c^T x \\
& Ax = b \\
& x \in \mathbb{R}_{\geq 0}^n \\
(P_4) \max_{\text{s.t.}} & b^T x \\
& A^T x = c \\
& x \in \mathbb{R}^n
\end{array}$$

Let z_1, z_2, z_3 and z_4 be the optimal objective function values of $(P_1), (P_2), (P_3)$ and (P_4) respectively.

- Determine the corresponding dual linear program for each of the above programs. Identify primal-dual pairs among the linear programs $(P_1) - (P_4)$.
- Show that if $(P_1), (P_2), (P_3), (P_4)$ are feasible, then $z_1 = z_2 = z_3 = z_4$.
- If $z_1 > z_2$, which linear program cannot be feasible? Explain your answer.

Solution

- First, we determine the dual for each linear program:

$$\begin{array}{ll}
(P_1) \max_{\text{s.t.}} & c^T x \\
& Ax = b \\
& x \in \mathbb{R}_{\geq 0}^n \\
(P_2) \min_{\text{s.t.}} & c^T x \\
& Ax = b \\
& x \in \mathbb{R}_{\geq 0}^n \\
(P_3) \min_{\text{s.t.}} & b^T x \\
& A^T x \geq c \\
& x \in \mathbb{R}^n \\
(P_4) \max_{\text{s.t.}} & b^T x \\
& A^T x = c \\
& x \in \mathbb{R}^n \\
(D_1) \min_{\text{s.t.}} & b^T y \\
& A^T y \geq c \\
& y \in \mathbb{R}^n \\
(D_2) \max_{\text{s.t.}} & b^T y \\
& A^T y \leq c \\
& y \in \mathbb{R}^n \\
(D_3) \max_{\text{s.t.}} & c^T y \\
& Ay = b \\
& y \in \mathbb{R}_{\geq 0}^n \\
(D_4) \min_{\text{s.t.}} & c^T y \\
& Ay = b \\
& y \in \mathbb{R}^n
\end{array}$$

We observe that $(P_1) - (P_3)$ is the only primal-dual pair.

b) If all LPs are feasible, then we get:

$$z_3 \geq z_1 \geq z_2 \geq z_4 \geq z_3,$$

which implies $z_3 = z_1 = z_2 = z_4$.

- The first inequality ($z_3 \geq z_1$) follows from weak duality, because (P_1) and (P_3) are a primal-dual pair, i.e., $(P_3) \equiv (D_1)$ and feasible.
- The second inequality ($z_1 \geq z_2$) follows because (P_1) is a maximization problem and (P_2) is a minimization problem and both have the same feasible set.
- To show the third inequality ($z_2 \geq z_4$) we need two steps.
First we observe that due to weak duality, the objective value of (P_2) is greater or equal to the objective value of its dual (D_2) .
And second, if we compare (D_2) and (P_4) , we see that they only differ in the feasible region. Since the feasible region of (P_4) is contained in the feasible region of (D_2) , i.e., $\{x \in \mathbb{R}^n : A^T x = c\} \subseteq \{y \in \mathbb{R}^n : A^T y \leq c\}$, the optimal objective function value of (D_2) has to be greater or equal than the one of (P_4) .
Combining these two steps we get $z_2 = \text{OPT}(P_2) \geq \text{OPT}(D_2) \geq \text{OPT}(P_4) = z_4$.

- Finally, the last inequality ($z_4 \geq z_3$) follows because (P_3) is a minimization problem, has the same objective function as (P_4) and its feasible set contains that of (P_4) .
- c)
- $(P_1) - (P_3)$ is a primal-dual pair. From strong duality it follows that $z_1 = z_3$ and thus (P_3) is feasible.
 - From part b) we know that if $(P_2), (P_3), (P_4)$ are feasible then $z_2 \geq z_4$ and $z_4 \geq z_3$.
 - It follows that (P_4) cannot be feasible because otherwise $z_2 \geq z_1$ should hold.

Exercise T4.3. Rock-Paper-Scissors

Problem

George and Max play rock-paper-scissors. The rules are simple: both players simultaneously choose a symbol from the set {Scissors, Rock, Paper}. Scissors beats paper, paper beats rock and rock beats scissors. If both players choose the same symbol, the round ends in a draw. The winner of each round receives 2 euros from the loser. In the case of a draw, no money is exchanged.

- Build the payoff matrix of this game for Max and George.
- Model the problem of finding an optimal strategy for George and Max as a linear program. State the optimal strategy and the win/loss for both players.
- Max proposes the following change to George. In the event of a tie, George receives 1.5 euros from Max if rock or scissors was played. If paper was played, Max receives 3 euros from George. Determine the optimal strategies for this modified problem. Should George accept the proposal? Justify your answer.

Hint: Use Gurobi to solve the LPs.

Solution

- The payoff matrix results from all possible combinations:

G\M	S	R	P	Var
S	0	-2	2	x_1
R	2	0	-2	x_2
P	-2	2	0	x_3
Var	y_1	y_2	y_3	

- George's optimization problem:

$$\begin{array}{ll}
 (P) \text{ Max} & g \\
 \text{s.t.} & \\
 & 0x_1 + 2x_2 - 2x_3 \geq g \\
 & -2x_1 + 0x_2 + 2x_3 \geq g \\
 & 2x_1 - 2x_2 + 0x_3 \geq g \\
 & x_1 + x_2 + x_3 = 1 \\
 & x_1, x_2, x_3 \geq 0 \\
 & g \in \mathbb{R}
 \end{array}
 \qquad
 \begin{array}{ll}
 (P_{NF}) \text{ Max} & g^+ - g^- = g \\
 \text{s.t.} & \\
 & 0x_1 - 2x_2 + 2x_3 + g^+ - g^- \leq 0 \\
 & 2x_1 + 0x_2 - 2x_3 + g^+ - g^- \leq 0 \\
 & -2x_1 + 2x_2 + 0x_3 + g^+ - g^- \leq 0 \\
 & -x_1 - x_2 - x_3 \leq -1 \\
 & x_1 + x_2 + x_3 \leq 1 \\
 & x_1, x_2, x_3, g^+, g^- \geq 0
 \end{array}$$

Max's optimization problem:

$$\begin{array}{ll}
 (D_{NF}) \text{ Min} & -\pi^- + \pi^+ = \pi \\
 \text{s.t.} & \\
 & 0y_1 + 2y_2 - 2y_3 - \pi^- + \pi^+ \geq 0 \\
 & -2y_1 + 0y_2 + 2y_3 - \pi^- + \pi^+ \geq 0 \\
 & +2y_1 - 2y_2 + 0y_3 - \pi^- + \pi^+ \geq 0 \\
 & y_1 + y_2 + y_3 \geq 1 \\
 & -y_1 - y_2 - y_3 \geq -1 \\
 & y_1, y_2, y_3, \pi^-, \pi^+ \geq 0
 \end{array}$$

Optimal strategy:

$$x = y = (1/3, 1/3, 1/3)$$

$$g = \pi = 0$$

- Modified problem:

G\M	Sch	St	P	Var
Sch	1.5	-2	2	x_1
St	2	1.5	-2	x_2
P	-2	2	-3	x_3
Var	y_1	y_2	y_3	

$$\begin{aligned}
(P_{NF}) \text{ Max } & g^+ - g^- = g \\
\text{s.t.} & \\
-1.5x_1 - 2x_2 + 2x_3 + g^+ - g^- & \leq 0 \\
2x_1 - 1.5x_2 - 2x_3 + g^+ - g^- & \leq 0 \\
-2x_1 + 2x_2 + 3x_3 + g^+ - g^- & \leq 0 \\
-x_1 - x_2 - x_3 & \leq -1 \\
x_1 + x_2 + x_3 & \leq 1 \\
x_1, x_2, x_3, g^+, g^- & \geq 0
\end{aligned}$$

Optimal Strategy: $x = (0.4667, 0.5333, 0)$, $y = (0, 0.5333, 0.4667)$, $g = \pi = -0.13$. This means that Max wins in expectation 0.13 euros per game, which George loses. Therefore, George should not agree to this proposal.

Self-study Exercises

Exercise S4.1. Duality

Problem

- (P)
$$\begin{array}{ll} \text{Max} & x_1 + x_2 \\ \text{s.t.} & -3x_1 + 2x_2 \leq -1 \\ & x_1 - x_2 \leq 2 \\ & x_1, x_2 \geq 0 \end{array}$$
- a) Formulate the corresponding dual LP to (P)
- b) List all complementary slackness conditions of these LPs.
- c) Does the primal LP have feasible solutions? Justify your answer.
- d) Is the primal LP bounded? Justify your answer.

Solution

- a) The dual problem (D) is given by

$$\begin{array}{ll} (D) \text{ Min} & -y_1 + 2y_2 \\ \text{s.t.} & -3y_1 + y_2 \geq 1 \\ & 2y_1 - y_2 \geq 1 \\ & y_1, y_2 \geq 0. \end{array}$$

- b) From the complementary slackness conditions we get

$$\begin{array}{ll} y_1(-3x_1 + 2x_2 + 1) = 0 & x_1(-3y_1 + y_2 - 1) = 0 \\ y_2(x_1 - x_2 - 2) = 0 & x_2(2y_1 - y_2 - 1) = 0 \end{array}$$

- c) The primal LP has a feasible solution. It is sufficient to specify one such solution. For example, we set $x_2 = 0$. This results in $3x_1 \geq 1$ and $x_1 \leq 2$ for each of the two constraints. Therefore, $(x_1, x_2) = (1, 0)$ is a feasible solution.
- d) We know from the lecture that if the primal is unbounded, the dual is infeasible. **Note that the reverse does not hold in general.** But we know that if the dual is infeasible, the primal is either infeasible or unbounded. Since we observed in c) that the primal is not infeasible, it can only be unbounded if the dual is infeasible. Therefore, in this specific setting it suffices to show that the dual LP is infeasible: To show this, suppose $y = (y_1, y_2)$ satisfies the dual constraints. y must also satisfy the constraint obtained by summing up both constraints. This new constraint is $-y_1 \geq 2$, or $y_1 \leq -2$. However, this constraint contradicts the constraint $y_1 \geq 0$. Thus, the set of feasible dual solutions is empty and hence the primal LP is unbounded.

Exercise S4.2. Duality & Complementary Slackness

Problem

- a) Derive the dual LP (D) from the following LP:

$$\begin{aligned}
(P) \quad & \text{Max} \quad 15x_1 + 5x_2 - 15x_3 \\
& \text{s.t.} \quad \begin{aligned} -x_1 + x_2 &\leq 2 \\ x_1 - x_2 + 2x_3 &\leq 2 \\ 2x_1 - 2x_3 &\leq 2 \\ -x_1 + 4x_2 + x_3 &\geq -1 \\ x_1 &\geq 0 \\ x_3 &\leq 0 \end{aligned}
\end{aligned}$$

- b) State all complementary slackness conditions.
- c) Let $y^r = (9, 0, 16, -1)$ be a dual point. Use the complementary slackness conditions to check whether y is an optimal solution for (D) .
- d) Let $x^r = (1, 3, 0)$ be a primal point. Use the complementary slackness conditions to check whether x is an optimal solution for (P) .

Solution

a)

$$\begin{aligned}
(D) \quad & \text{Min} \quad 2y_1 + 2y_2 + 2y_3 - y_4 \\
& \text{s.t.} \quad \begin{aligned} -y_1 + y_2 + 2y_3 - y_4 &\geq 15 \\ y_1 - y_2 + 4y_4 &= 5 \\ 2y_2 - 2y_3 + y_4 &\leq -15 \\ y_1, y_2, y_3 &\geq 0 \\ y_4 &\leq 0 \end{aligned}
\end{aligned}$$

b) The complementary slackness conditions are given by

$$\begin{aligned}
(-x_1 + x_2 - 2)y_1 &= 0 & (-y_1 + y_2 + 2y_3 - y_4 - 15)x_1 &= 0 \\
(x_1 - x_2 + 2x_3 - 2)y_2 &= 0 & (y_1 - y_2 + 4y_4 - 5)x_2 &= 0 \\
(2x_1 - 2x_3 - 2)y_3 &= 0 & (2y_2 - 2y_3 + y_4 + 15)x_3 &= 0 \\
(-x_1 + 4x_2 + x_3 + 1)y_4 &= 0
\end{aligned}$$

- c) First, we check if y is feasible by plugging it into (D) . We get $[24 \geq 15], [5 = 5], [-33 \leq -15], [9, 0, 16 \geq 0], [-1 \leq 0]$. Thus y satisfies all dual constraints and is therefore a feasible point in (D) .

Since the first and third constraints have slack we get $x_1 = x_3 = 0$. Since $y_1, y_3, y_4 \neq 0$ the point x has to satisfy

$$(-x_1 + x_2 - 2) = (2x_1 - 2x_3 - 2) = (-x_1 + 4x_2 + x_3 + 1) = 0.$$

We know that $x_1 = 0$, which means that $x_2 = 2$ (first condition). This gives us the primal point $x^T = (0, 2, 0)$. But, if we look at the third condition, i.e., $(2x_1 - 2x_3 - 2) = 0$ we get $-2 \neq 0$, which means that there exists no point x satisfying all complementary slackness conditions. This means that y cannot be an optimal solution.

- d) First, we check if x is feasible by plugging it into (P) . We get $[2 \leq 2], [-2 \leq 2], [2 \leq 2], [11 \geq -1], [1 \geq 0], [0 \leq 0]$. Thus x satisfies all primal constraints and is therefore a feasible point in (D) .

Since the second and fourth primal constraints have slack, we get $y_2 = y_4 = 0$. Since $x_1, x_2 \neq 0$ the point y has to satisfy

$$(-y_1 + y_2 + 2y_3 - y_4 - 15) = (y_1 - y_2 + 4y_4 - 5) = 0.$$

this gives us $y_1 = 5$ and $y_3 = 10$, i.e., $y = (5, 0, 10, 0)^T$. Finally, we check if y is feasible, i.e., if it satisfies all dual constraints: $[15 \geq 15], [5 = 5], [-20 \leq -15], [5, 0, 10 \geq 0], [0 \leq 0]$. From the complementary slackness theorem it follows that x is an optimal solution for (P)